

Prelab 2.1 More Raspberry Pi configuration and package installation

Learning Goals

Students will be able to:

1. Configure the basic parameters to read sensors through Raspberry Pi
2. Calculate time domain statistics for datasets using Python
3. Transform time-domain data using fast fourier transform in Python
4. Plot time-domain and frequency domain datasets using Python

1.1 Introduction

In Lab2, we will use a tri-axis MEMS (Micro-Electromechanical System) based accelerometer, ADXL345 module (<https://www.analog.com/en/products/adxl345.html>, available on Jan. 25, 2023 and the data sheet is uploaded on Brightspace), to measure accelerations of machinery. The ADXL345 is a cheap and low-power accelerometer with a high resolution (13-bit) measurement at up to ± 16 g. Measurement is performed by digital output data formatted as 16bit and accessible through either an SPI (Serial Peripheral Interface) or I2C (or I2C, IIC, Inter-Integrated Circuit) digital interface. We will use the I2C method to access data of the sensor using Adafruit ADXL Python package (<https://circuitpython.readthedocs.io/projects/adxl34x/en/latest/>, available on Jan. 25, 2023). Also, signal processing of collected data is one of the major keys in data analytics.

Prelab 2 is to prepare to use the ADXL345 sensor module and signal processing with visualization of signal in Raspberry Pi. For the signal processing, we will analyze data in both time-domain and frequency-domain.

1.2 Raspberry Pi configuration to enable I²C interface

To access data from ADXL345, we will use I2C interface of Raspberry Pi. Please note, **part of this was completed in Prelab 1**. The steps for enabling the I2C interface are described in GUI (VNC access). If you want to do it in CLI, please use `sudo raspi-config`. Continue to follow the procedure below for the remainder.

1. Click 'applications menu' (Pi logo on the upper left side)

2. Click 'Preferences'
3. Click 'Control center'
4. Click 'Interfaces' tab
5. Click 'Enable' of 'I2C' as Figure 1.
6. Reboot Raspberry Pi

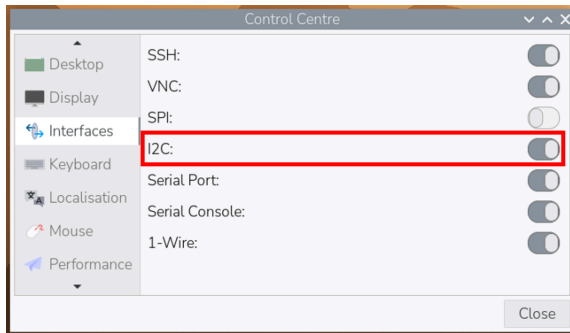


Figure 1 Enabling I2C (1-wire can be enabled)

Make sure Raspberry Pi up to date. And Install I2C package of Raspberry Pi.

Raspberry Pi - Terminal

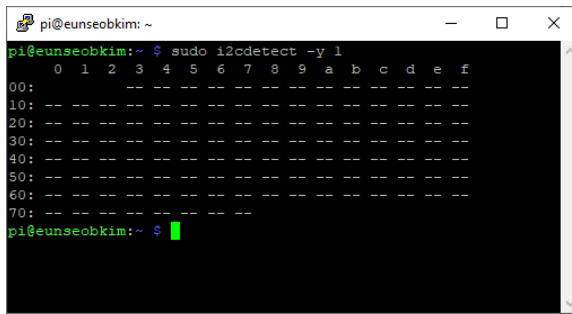
```
sudo apt update
sudo apt upgrade
sudo apt install i2c-tools -y
```

To check the I2C interface, you can use the command 'sudo i2cdetect -y 1' as below. Because there is no I2C interface connected now, the result will look like Figure 2. If you are interested in I2C communication protocol in detail, visit and go through Sparkfun tutorial for I2C (<https://learn.sparkfun.com/tutorials/i2c/all>, available on Jan. 20, 2022).

Raspberry Pi - Terminal

```
sudo i2cdetect -y 1
```

- If you have error and your Raspberry Pi is the very first version of Raspberry Pi (Raspberry Pi model B), 'sudo i2cdetect -y 0' will work.
-



```
pi@eunseobkim: ~  
pi@eunseobkim:~$ sudo i2cdetect -y 1  
    0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f  
00:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
10:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
20:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
30:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
40:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
50:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
60:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
70:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
pi@eunseobkim:~$
```

Figure 2 Connected I2C interface check

Task 1.1

Capture your terminal window just like Figure 2 after checking I2C interface detection and attach to the report below:



```
pi@robertclaud:~$ sudo i2cdetect -y 1  
    0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f  
00:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
10:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
20:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
30:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
40:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
50:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
60:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
70:  --  --  --  --  --  --  --  --  --  --  --  --  --  --  --  
pi@robertclaud:~$
```

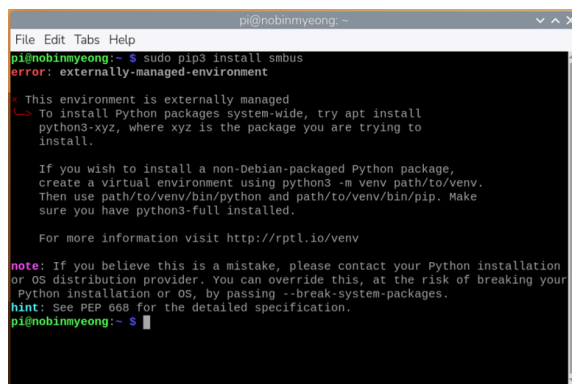
1.3 Installation of related Python packages

To install the related Python packages for ADXL345 sensor and signal processing, run the installation command lines below. For signal processing, we will mainly use NumPy (<https://numpy.org/>) and SciPy (<https://scipy.org/>) libraries. Matplotlib (<https://matplotlib.org/>) is a powerful library to visualize data in Python. Both NumPy and SciPy are the fundamental Python libraries for scientific data analytics. Those are frequently used with machine learning applications as well. It is highly recommended to get familiar with those packages by trying tutorials available on the Internet.

```
sudo pip3 install smbus
sudo pip3 install adafruit-circuitpython-ADXL34x
# Below are additional packages for signal processing and
visualization.
# Because wheels and setup.py errors when installing scipy,
apt command is used for scipy and matplotlib installations.
sudo apt install python3-numpy
sudo apt install python3-scipy
sudo apt install python3-matplotlib
# If error occurs when importing scipy, run the command
below.
sudo apt install libatlas-base-dev
```

If you see this error, please add "`--break-system-packages`" after command.

(e.g. `sudo pip3 install smbus --break-system-packages`)



Task 1.2

Run the command, `pip3 list`, and capture a screenshot of your terminal window just like Figure 3 below to see the installed ADXL34X package

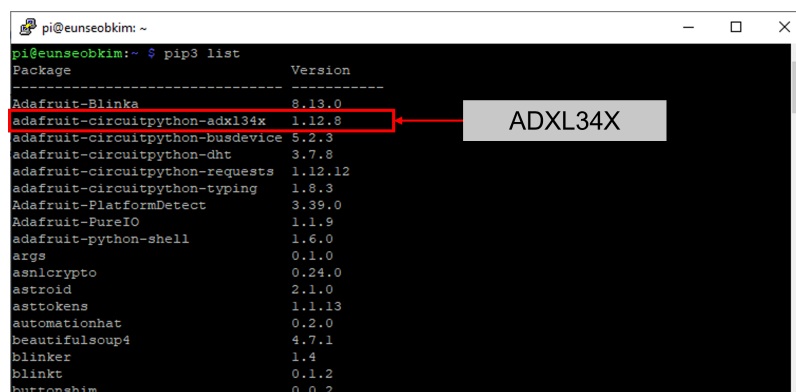
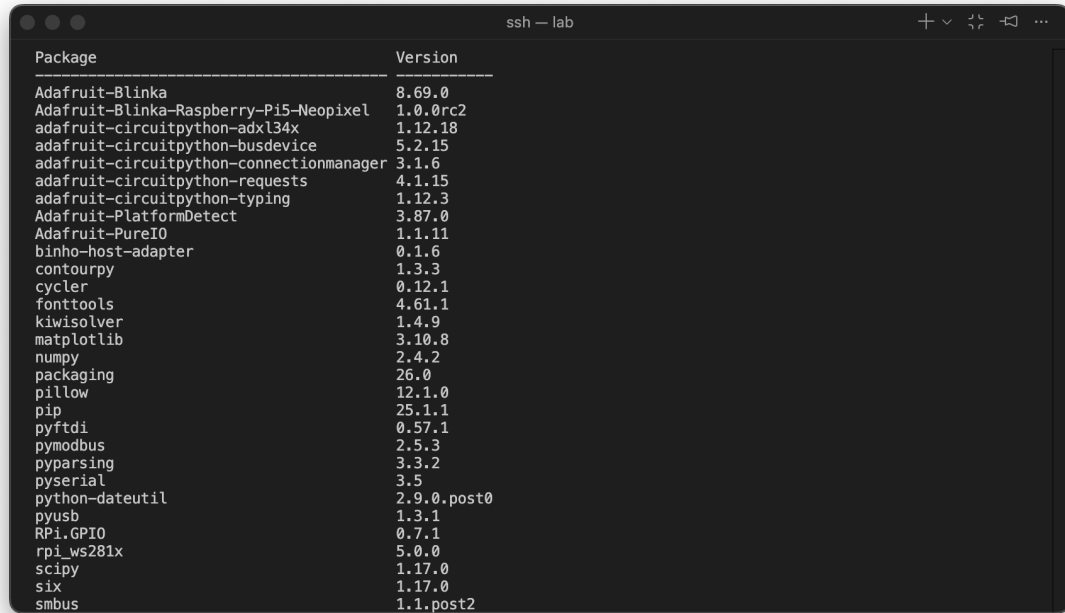


Figure 3 Terminal window to check the installed Python packages

A terminal window titled 'ssh - lab' with standard window controls. It displays a list of installed Python packages and their versions in a table format.

Package	Version
Adafruit-Blinka	8.69.0
Adafruit-Blinka-Raspberry-Pi5-Neopixel	1.0.0rc2
adafruit-circuitpython-adxl34x	1.12.18
adafruit-circuitpython-busdevice	5.2.15
adafruit-circuitpython-connectionmanager	3.1.6
adafruit-circuitpython-requests	4.1.15
adafruit-circuitpython-typing	1.12.3
Adafruit-PlatformDetect	3.87.0
Adafruit-PureIO	1.1.11
binho-host-adapter	0.1.6
contourpy	1.3.3
cycler	0.12.1
fonttools	4.61.1
kiwisolver	1.4.9
matplotlib	3.10.8
numpy	2.4.2
packaging	26.0
pillow	12.1.0
pip	25.1.1
pyftdi	0.57.1
pymodbus	2.5.3
pyparsing	3.3.2
pyserial	3.5
python-dateutil	2.9.0.post0
pyusb	1.3.1
RPi.GPIO	0.7.1
rpi_ws281x	5.0.0
scipy	1.17.0
six	1.17.0
smbus	1.1.post2

Please continue to [Prelab 2.2 here](#).